Unusual uses of Ansible Automation Platform and Event-Driven Ansible

Rune Juhl
Open source architect, Atea
rune.juhl@atea.dk

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Who're we?

- Atea Denmark has a department called Open Source
- We work with… open source!
  - Linux (all flavors)
  - PostgreSQL, MariaDB, Redis
  - Kubernetes, Docker, Podman
  - Ansible, Puppet
  - DevOps, GitOps, et al.

Who am I?

- Rune Juhl
- Architect, Open Source, Atea Denmark
- Linux user for > 22 years
- Rabid advocate of open-source software
- Emacs user

What is this about?

- Playing around with Ansible and EDA because
  - It's fun!
  - It's educational!
  - ...because why not?
- Beware!
  - This is exploration, not guaranteed to work
  - Not polished
  - No AI

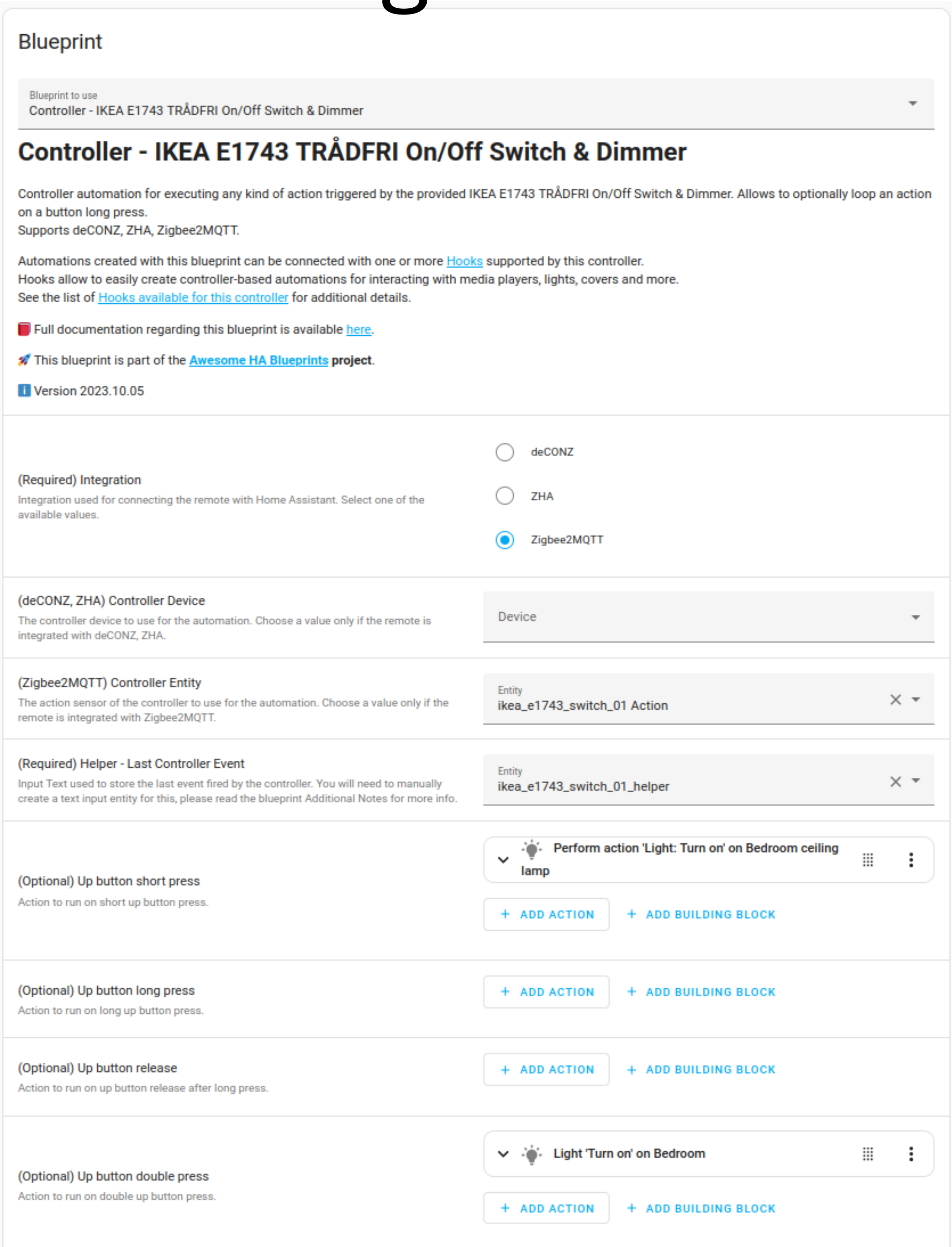
Why home automation?

- Been using Home Assistant for a long time
- "Programming" in YAML is not fun
- CLI >> GUI
- ...because I felt like it!

Link to "Why the fuck are we templating yaml?", a blog post by Lee Briggs

https://leebriggs.co.uk/blog/2019/02/07/why-are-we-templating-yaml

GUI fatigue



YAML fatigue

```yaml
condition:
  # if triggered by the homeassistant platform check if the automation should be run, as specified by the user
  # if triggered by something else, just continue
  - condition: and
    conditions:
      # if triggered by the run time, check that the automation is allowed to run today
      - >-
        {%- set today = now().weekday() -%} {{ trigger.id != "run_time" or today == 0 and day_monday or today == 1 and day_tuesday or
      # if triggered due to home assistant startup, check that the feature is enabled
      - '{{ trigger.id != "homeassistant_start" or trigger_at_homeassistant_startup }}'
      # if triggered due to home assistant startup or to a custom trigger event, perform the execution check
      # check whether the automation run at or after the expected run time
      # if this is the case, there's no need to run the automation again
      # else, the automation was not able to run at the expected time, and was not executed after it, hence run it now
      # if block late execution is enabled, check also that no more than the provided max elapsed time has passed since the expected r
      - >-
        {%- set cdt = now() | as_timestamp | timestamp_local | as_datetime -%}
        {%- set lsrdt = ((states(helper_storage) | from_json).last_triggered if (states(helper_storage) | regex_match(json_regex)) els
        {%- set idt = strptime(states(time_entity),time_fmt) -%}
        {%- set days = [day_monday, day_tuesday, day_wednesday, day_thursday, day_friday, day_saturday, day_sunday] -%}
        {%- set cwd = cdt.weekday() -%}

        {% macro day_offset(day) -%}
          {%- if days[day] -%}
            {{ 7 if cwd == day and cdt.time() < idt.time() else (cwd-day)%7 }}
          {%- else -%}
            -1
          {%- endif -%}
        {%- endmacro %}

        {%- set weekdays = [day_offset(0)|int, day_offset(1)|int, day_offset(2)|int, day_offset(3)|int, day_offset(4)|int, day_offset(

        {%- if weekdays | length > 0 -%}
          {%- set closest_weekday = weekdays | min -%}
          {%- set day_diff = timedelta(weeks=weeks_frequency-1, days=closest_weekday) -%}
          {%- set edt = (cdt - day_diff - timedelta(hours=cdt.hour, minutes=cdt.minute, seconds = cdt.second) + timedelta(hours=idt.ho
          {{ trigger.id not in ["homeassistant_start","custom_trigger_event"] or lsrdt < edt and (not block_late_execution or edt >= c
        {%- else -%}
          {{ false }}
        {%- endif -%}
action:
  # store the current datetime into the helper, since the automation was successfully triggered
  - service: input_text.set_value
    data:
      entity_id: !input helper_storage
```

Let's replace it!

- ...with Ansible and more YAML!
- ...maybe more flexible?
  - No restrictions on programming languages
  - Lots of Ansible modules
- How hard can it be?
- ...he said, before spending countless hours debugging and monkey-patching EDA Decision Environments


ME
YAML
MORE YAML
imgflip.com

Home automation with Ansible and EDA!

Trigger sources

- MQTT
  - Zigbee2MQTT
  - DNS server logs
- HTTP server-sent events (SSE)
  - ESPHome devices
- Websocket
  - Home Assistant
- Polling HTTP

Respond using Ansible

- MQTT
  - Home Assistant
- HTTP REST API
  - UniFi Network Application (undocumented?)
  - Home Assistant (slightly less undocumented)
  - Drinks machine
- Anything you can think of

Ideas

- Listen to presence sensor, turn on lights
- Throttle Wi-Fi if the kids play Roblox at odd hours
- Detect loo roll empty (TP LOAD LETTER)
- Hook up drinks machine and have it dispense drinks when you get home
- Ingest events from mm-Wave radar, turning on smoke machine when Halloweeners nears the house

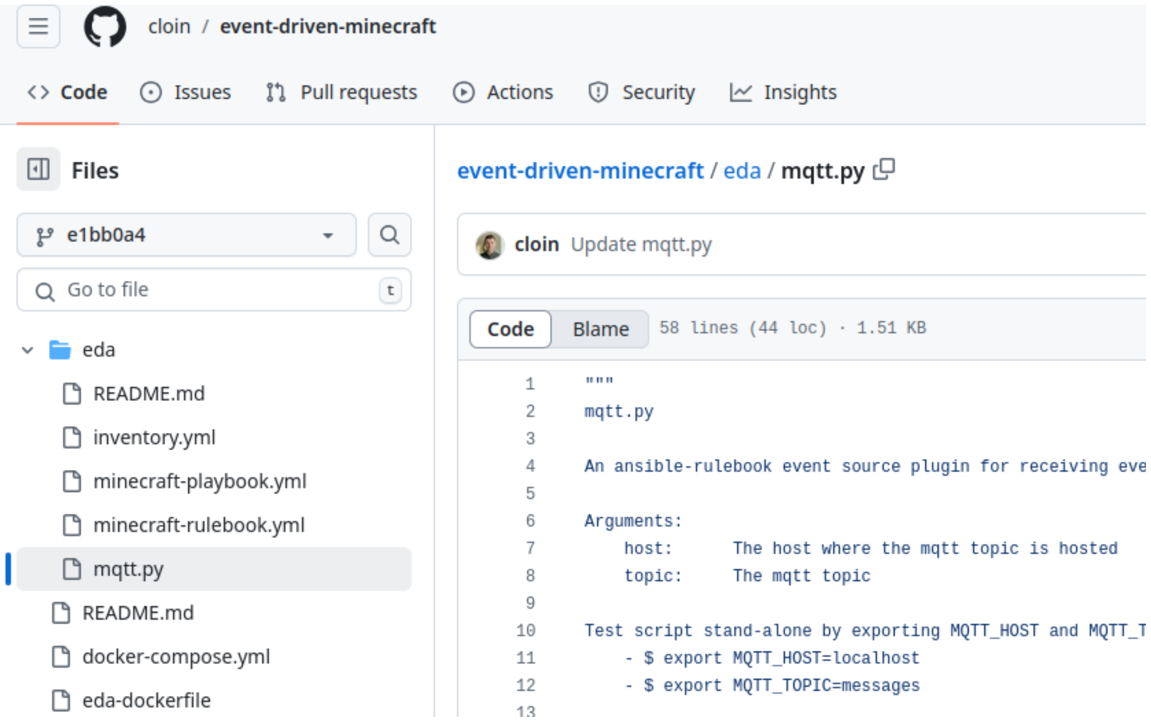
Throttling the kids

Imagine the following:

- Kids play Roblox/Fortnite/Among Us
- They become addicted and can't stop
- Let's limit Wi-Fi access EDA!

EDA and MQTT

- EDA uses event source plugins
- ...MQTT not supported by default
- Thank you, Colin McNaughton!
- Only needed a few changes



Pipeline: Logs

```
Oct 28 17:19:49 dnsmasq[198080]: query[A] clientsettingscdn.roblox.com from 192.168.1.226
Oct 28 17:19:49 dnsmasq[198080]: forwarded clientsettingscdn.roblox.com to 208.67.222.222
Oct 28 17:19:49 dnsmasq[198080]: reply clientsettingscdn.roblox.com is <CNAME>
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.84
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.94
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.16
Oct 28 17:19:49 dnsmasq[198080]: reply d2v57ias1m20gl.cloudfront.net is 18.173.5.77
```

Pipeline: Logstash

```
input {
  file {
    path => ['/var/log/pihole.log']
    mode => 'tail'
    start_position => 'end'
  }
}
filter {
  mutate { gsub => ["message", "^.+?: ", ""] }
  if ([message] !~ "^query") { drop {} }
  dissect {
    mapping => { message => "query[%{type}] %{name} from %{source}" }
  }
  prune { whitelist_names => ["^type$", "^name$", "^source$"] }
  if ([source] =~ "^(127\.|172\.)") { drop {} }
}
output {
  stdout { codec => 'json' }
  mqtt {
    host => "192.168.1.243"
    port => 1883
    topic => "dns"
  }
}
```

Pipeline: EDA

```yaml
- name: mqtt events
  hosts: localhost
  sources:
    - runejuhl.weird_ansible.mqtt:
        host: '{{ mqtt_server }}'
        topic: 'dns'

  rules:
    - name: Run job on DNS lookup
      condition:
        all:
          - event.payload.name == "clientsettingscdn.roblox.com"
          # match 21, 22, 23, 00, 01, 02, 03, 04, 05, 06, 07
          - event.meta.received_at is regex('T(2[123]|0[0-7]):')
      action:
        run_job_template:
          name: unifi_throttle
          organization: Default
          job_args:
            extra_vars:
              throttle_client: true
```

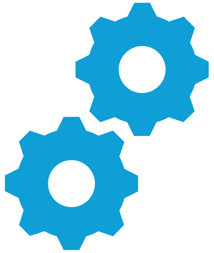
Pipeline: UniFi

```yaml
- name: |-
    Throttle wifi
  hosts: localhost
  gather_facts: false
  tasks:

  - name: Login
    ansible.builtin.uri:
      url: '{{ unifi_baseurl }}/api/login'
      method: POST
      headers:
        content-type: 'application/json'
      return_content: true
      body:
        {
          "username":"{{ unifi_username }}",
          "password":"{{ unifi_password }}",
          "remember":false,
          "strict":true
        }
      body_format: 'json'
    register: login

  - name: Get devices
    ansible.builtin.uri:
      url: '{{ unifi_baseurl }}/v2/api/site/default/clients/active?includeTrafficUsage=true&includeUnifiDevices=true'
      headers:
        Cookie: |-
          {{ login.cookies_string }}
        X-Csrf-Token: |-
          {{ login.cookies.csrf_token }}
      return_content: true
    register: devices

  - ansible.builtin.set_fact:
      device: |-
        {{ devices.json|selectattr('ip', 'eq', ansible_eda.event.payload.source)|first  }}

  - name: Throttle device
    ansible.builtin.uri:
      url: '{{ unifi_baseurl }}/api/s/default/rest/user/{{ device.user_id }}'
      method: PUT
      headers:
        Cookie: |-
          {{ login.cookies_string }}
        X-Csrf-Token: |-
          {{ login.cookies.csrf_token }}
      return_content: true
      body:
        usergroup_id: |-
          {% if throttle_client %}5ff6a0c246e0fb01138ed94f{% endif %}
      body_format: 'json'
    register: call_throttle

  - ansible.builtin.debug:
      msg: |
        {{ call_throttle.json }}
    when: |-
      not call_throttle.json.meta.rc == "ok"
```

Loo roll detector

Common household issue

Not fun being left out to dry



A case for automation!

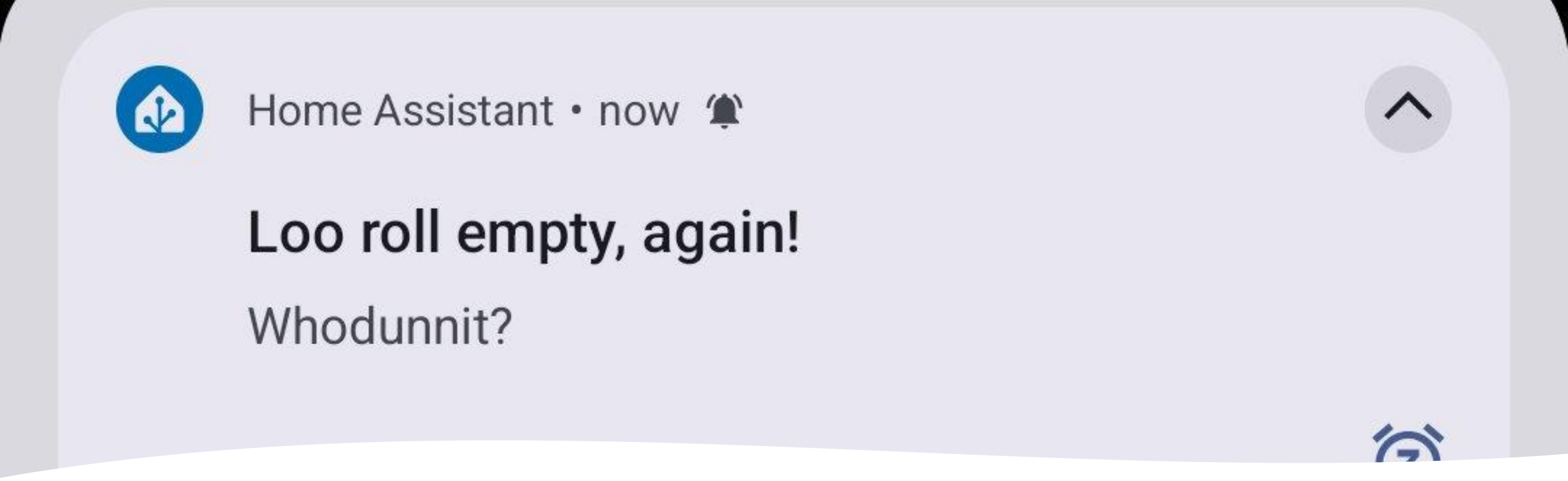
Technical specs

- NodeMCU dev board (ESP8266)
- TCS34725 RGB sensor
- Li-Ion 18650 battery holder

Pipeline: Server-Sent Events

- Event plugin based on MQTT plugin
- Added matching on event IDs

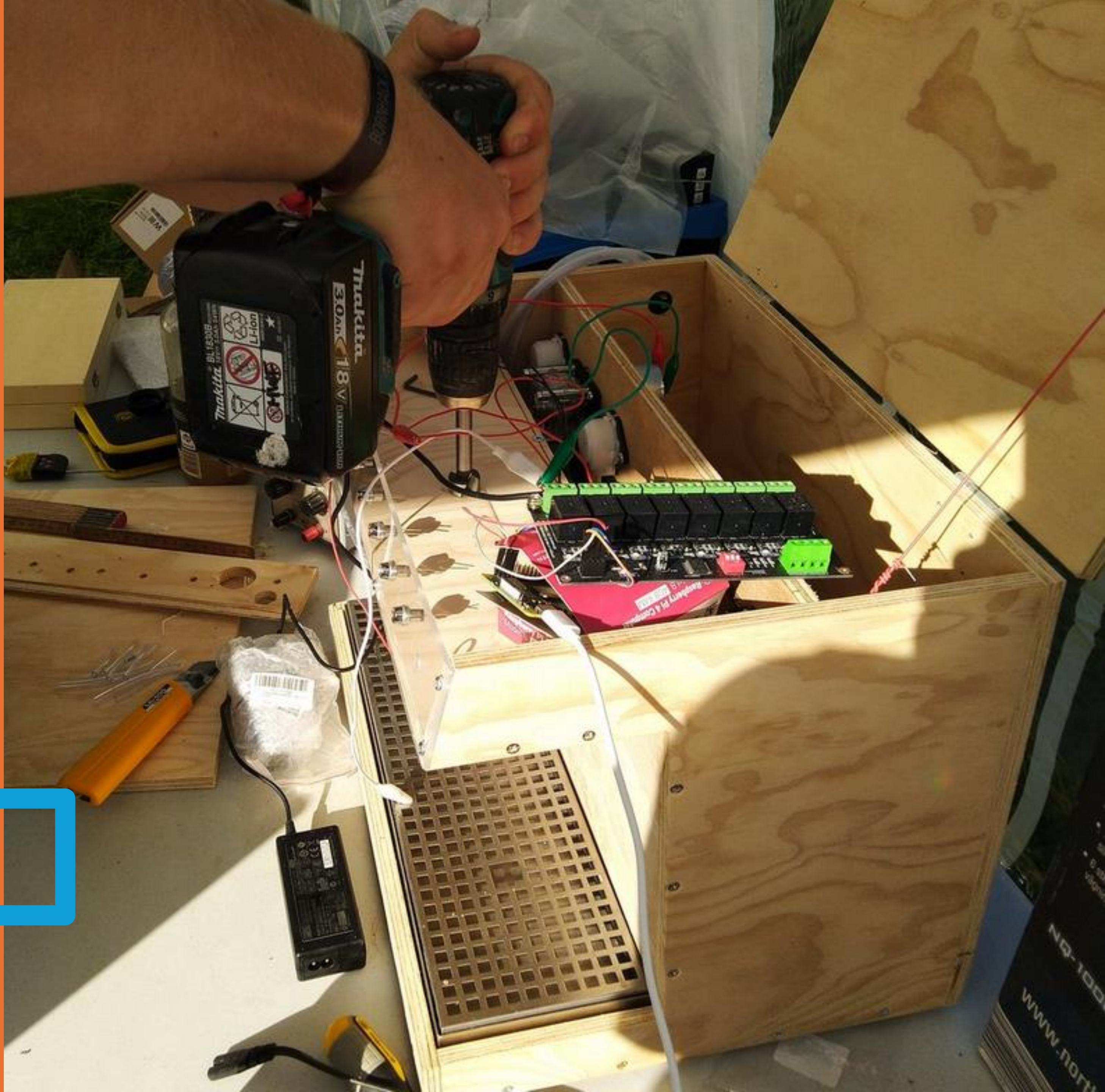
```yaml
- name: http sse events
  hosts: localhost
  default_events_ttl: "2 seconds"
  match_multiple_rules: false
  sources:
  - http_sse:
      url: '{{ loo_roll_basename }}/events'
      event_ids:
      - sensor-tcs34725_color_temperature
      - sensor-tcs34725_illuminance
```



Improvements

- ESP32-based presence detector?
  - Bluetooth MACs aren't stable
- Facial recognition using Frigate and Double Take?
- ...maybe an accessible stock is easier.

Drinks machine



Drinks machine

Built as a project for BornHack 2021

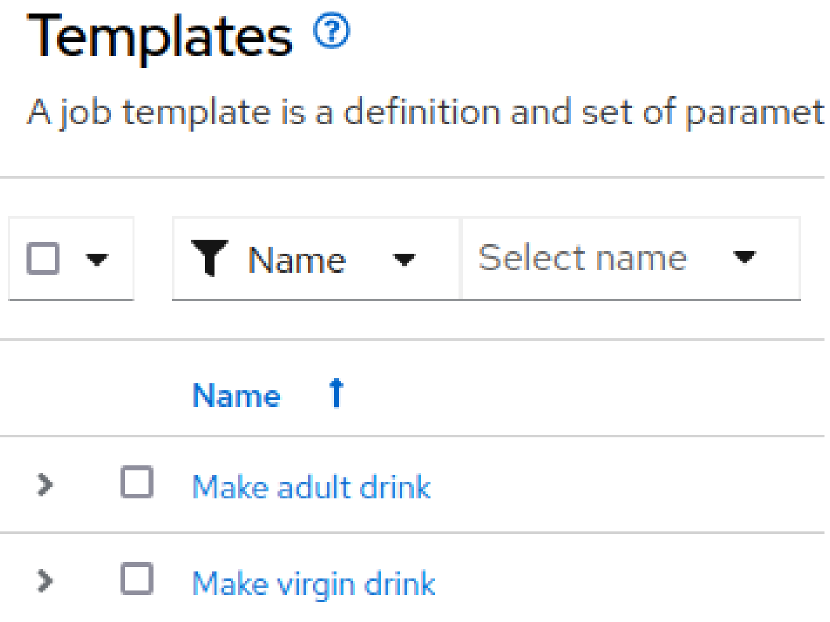
- Raspberry Pi
- I^2C relays
- 10 peristaltic pumps
- Backend in Clojure, web frontend in PowerShell

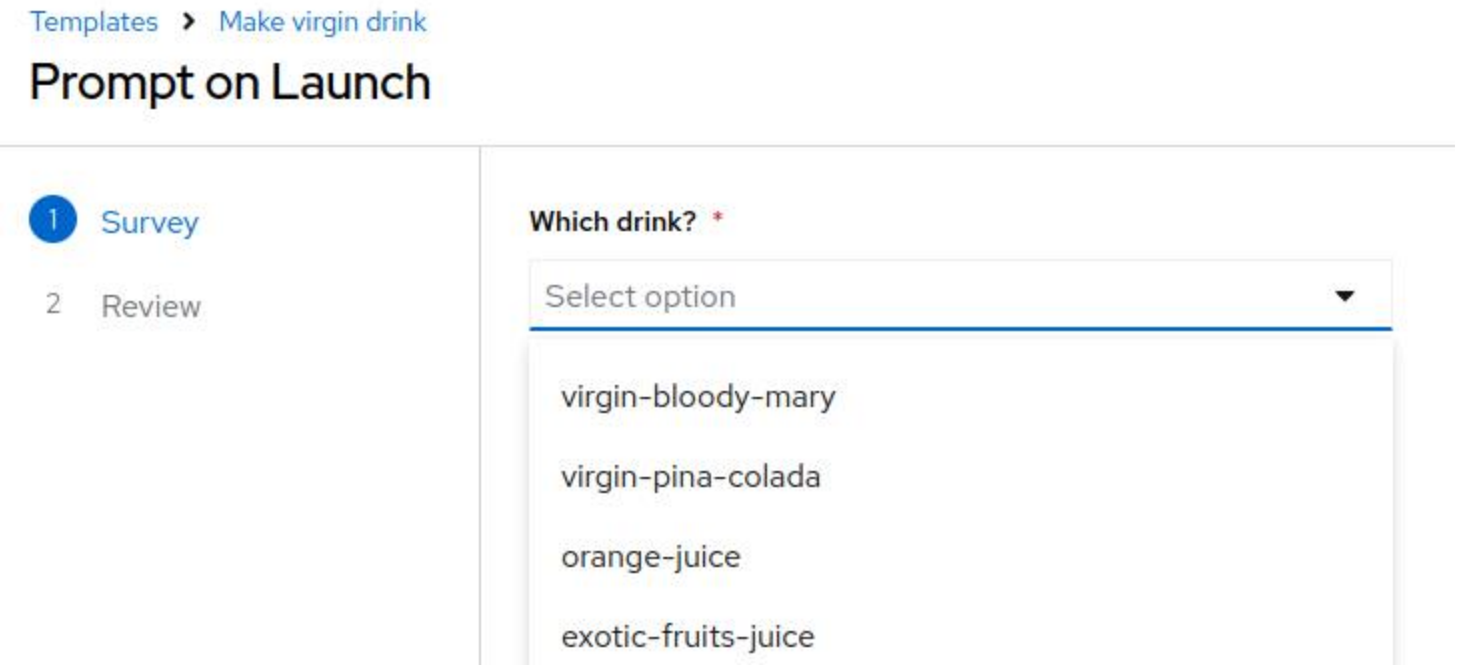
Status

- Almost worked at the time
- In storage ever since
  - ...much to the dismay of my kids
  - Until now!
- Fixed the last bug (I hope)
- Simple HTTP API (just POST, no body)

A private bartender

- Job templates allow us to limit access
  - Kids only allowed virgin cocktails
- Home Assistant REST API shows when I'm home
- EDA has `url_check` event plugin
  - Minor patch to allow authorization header





Troubles along the way

- AAP 2.5 containerized all-in-one mismatch with API URL in job template runner
- Decision Environment builds take *a lot* of disk space...
- Differences between ansible-rulebook and AAP EDA
- ...but I did what I set out to do!

That's it!

If the drinks machine worked during presentation, you're welcome to come up and have a (probably terrible) drink!